
EDUCATION

University of Waterloo

Bachelor of Computer Science

Waterloo, ON

2024 – Present

AWARDS AND HONORS

Cybersecurity Competition Participant (picoCTF) Global Top 1.5% (2023), **USA Computing Olympiad (USACO) Gold Division** (2023), **Canadian Computing Competition (CCC) Honor Roll** (Top 5%, 2023), **HiMCM Successful Participant** (2023), **AP Scholar with Distinction** (2023, 2024)

SKILLS

- **Programming:** Python, Java, JavaScript, SQL, C, C++, Racket
- **Frontend:** React, Vite, HTML, CSS, Responsive UI
- **Backend and Data:** Flask, Supabase, Data Processing
- **Tools and Platforms:** n8n, WordPress, Figma, Git, Linux, Visual Studio Code, npm, ESLint

PROJECTS

Bee Population Modeling and Hive Placement Optimization

C++, Python

Algorithm Design and Data Modeling

2025

- Built a simulation-based optimization project to model bee population trends and support hive placement decisions under environmental and spatial constraints.
- Implemented a **C++** search algorithm using DFS-style backtracking to evaluate valid hive placements and improve coverage under placement rules.
- Used **Python** to process ecological data, generate model inputs, and compare predicted outcomes across different placement strategies.

Helldivers Interactive Web System

React, Vite, JavaScript, HTML, CSS

Core Frontend Development

Aug 2025 – Present

- Collaborated as a core developer on an interactive Helldivers-inspired web system, building a system-style interface with dynamic user interaction, game-like page flows, and reusable UI sections.
- Designed reusable **React** components for navigation sections, stratagem-style cards, interactive UI elements, layout systems, and state-driven user flows.
- Implemented character-driven interaction logic to replace simple click-based navigation with dynamic transitions, improving the overall user experience and project interactivity.
- Structured frontend code with modular component patterns, reusable CSS layouts, and organized state management to support future feature expansion and easier maintenance.
- Integrated **Vite**, npm scripts, and ESLint to improve local development workflow, code consistency, debugging speed, and maintainability.

Task Management REST API

Spring Boot, PostgreSQL, Docker

Backend Development

2025

- Developed RESTful APIs for creating, updating, deleting, and retrieving task records using **Spring Boot**.
- Integrated **PostgreSQL** for persistent storage and designed database tables for task status, descriptions, timestamps, and user records.
- Implemented centralized exception handling and structured API responses to make backend errors easier to debug and maintain.
- Containerized the application using **Docker** to simplify local setup and create a consistent development environment.

Card Game Implementation (Dou Di Zhu)

C++, Data Structures

Game Logic Design

2023

- Implemented core game logic including card representation, player turns, rule validation, and game state updates.
- Applied **C++** data structures to store hands, compare card combinations, and determine whether a player action followed game rules.
- Designed helper functions for hand ranking, turn validation, and win-condition checking to keep the game logic modular and easier to test.

EXPERIENCE

WE Digital Bootcamp

Waterloo, ON

Web Design and Digital Marketing Intern

May 2026 – Aug 2026

- Developed responsive, user-centered web solutions by applying UX/UI design principles, accessibility standards, and design thinking methodologies.
- Optimized digital presence through SEO, audience analysis, inbound marketing, content strategy, and web performance insights.
- Created marketing deliverables including website content, email campaigns, blogs, social media materials, and video marketing concepts.

Zhineng Education

Guangzhou, Guangdong

Computer Science Teaching Assistant

Sept 2023 – July 2024

- Instructed foundational software concepts including control flow, **OOP**, and algorithms in **Python** and **Java**.
- Resolved logical and runtime errors for **40+** students by implementing systematic debugging strategies and conducting code reviews.
- Architected **15+** custom programming exercises to reinforce modular programming and runtime efficiency.